

Yi Zhang

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Summary

Principal-level, hands-on security engineer with expertise in penetration testing, secure software development lifecycle (SSDLC), and AI/ML security.

Over 20 years of experience, beginning with Linux automation and system integration, and extending across Android, IoT, cloud and data platforms, and AI agent-based systems. Recent focus includes AI security governance, implementation of guardrails, and operationalizing security policies through automation and code-level enforcement (e.g., vibecoding).

Recognized for defining security strategy and translating high-level policies into actionable, enforceable engineering practices that integrate seamlessly into development workflows.

Experience

2022 – Present Staff Security Engineer | *TheTradeDesk.com*

- **AI Security Strategy & Enforcement:** Defined AI security policies and governance for enterprise adoption, including RAG and agentic systems. Led security reviews of AI/ML pipelines and applications. Translated high-level policies into actionable developer guidance, including AI guardrails and secure coding practices integrated into the development lifecycle.
- **Application Security:** Established and led SSDLC processes, including architecture reviews, code reviews, penetration testing, and internal R&D training
- **Infrastructure Security:** Designed and implemented a centralized PKI system; provided consulting on IT and data center security
- **Cloud & Data Security:** Helped standardize cloud security practices and integration, including IAM, key management, Snowflake and Databricks data platform security, and Kubernetes (RBAC & PSP)
- **Regulatory & Compliance Security Engineering:** Interpreted legal and contractual requirements related to data processing constraints (e.g., PAFACA, DoD DSP 2025) into actionable security controls and enforceable policies, bridging legal, compliance, and engineering.

2013 – 2022 Security Engineer | *Amazon.com*

- Designed, implemented, and maintained a vulnerability management system for Amazon smart devices
- Conducted Android malware analysis and reverse engineering
- Provided security consulting, including architecture review, threat modeling, and security automation
- Managed hardware security lab for Android device and application penetration testing and ad-hoc analysis

2007 – 2013 Sr. System SDET | *RedHat.com*

- Designed, implemented, and executed Linux server test automation (LDAP, Apache, SELinux, IPA, etc.)

2004 – 2007 Quality Engineer | *AOL.com*

- Developed API test plans, automation scripts, and verified bugs for AOL Video and AOL Blog

Qualification

Security Engineering

- Internal consulting and education on security policies, standards, and guidelines
- Cross-functional team coordination, including release and remediation planning
- Application security, including architecture reviews, code reviews, and penetration testing
- Threat modeling for web, cloud-based, and Android applications
- CI/CD pipeline security, software supply chain security (SBOM), and vulnerability management
- Developer consulting on cryptographic libraries, secure design research, and solution recommendations

AL/ML Systems & Security

- Supervised ML application design and implementation
- Custom LLM-RAG development for internal security tools
- AI/ML security governance, including guardrail design and secure usage patterns for AI agent-based systems
- Development of secure AI coding practices and enforcement via automated rulesets (Cursor)

Legal, Regulation, and Compliance

- Translation of legal and contractual data processing requirements (e.g., PAFACA, DoD DSP 2025) into enforceable security controls and policies
- Strong understanding of international, federal, and state regulations, including GDPR, CCPA, HIPAA, as well as SEC and FTC regulatory frameworks and broader security compliance standards

Systems & Programming

- Linux & Android: deep understanding of security models and configurations
- Proficient in Python, Perl, Ruby (on Rails), Bash, JavaScript, Java; working knowledge of C, C++

Education

- **Master of Studies in Law** (Cybersecurity & Privacy Law), USC – 2024
- **MS in Cybersecurity** (with Data Science Certificate), UC Berkeley – 2020
- **BS in Computer Science**, San Jose State University – 2012

Reference

- **LinkedIn Profile:** <https://www.linkedin.com/in/yizhangid/>
- **Security Legal Case Study:** <https://securitylawcase.com/>